SIM1: Physics-Aligned Simulator as Zero-Shot Data Scaler in Deformable Worlds

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Research Questions

- How can simulation be grounded so that synthetic data becomes a reliable substrate for real-robot deployment in deformable manipulation?
- Can physics-aligned simulation achieve zero-shot transfer from synthetic training data to real-world deformable object manipulation?
- What is the data scaling efficiency of physics-grounded synthetic data compared to real-world demonstrations for deformable manipulation tasks?

Core Idea

Traditional sim-to-real approaches fail for deformable objects because they use rigid-body physics and lack geometric/dynamic alignment with reality. SIM1 solves this through a real-to-sim-to-real (R2S2R) pipeline that first digitizes real scenes with submillimeter precision, then calibrates physics parameters to match real cloth behavior, and finally generates diverse trajectories via diffusion models. The key insight is that simulation must be 'grounded' in physical reality through three alignments: geometric (accurate 3D scanning), dynamical (physics parameter calibration), and behavioral (human-like motion synthesis). This creates synthetic data that behaves identically to real data, enabling zero-shot transfer.

Key Formula

Strain Constraint

$$C(e) = ||e_i - e_j|| - (1+\xi)l_0 \leq 0$$

C(e) is constraint violation, e_i and e_j are vertex positions, xi is maximum stretch ratio, l_0 is rest length. Activates virtual elastic forces when edge deformation exceeds threshold to prevent unrealistic stretching in cloth simulation.

Strain Energy Formulation

`E_{strain}^{(n)}(e) = \begin{cases} \frac{1}{2}k^{(n)}C(e)^2 + \lambda^{(n)}C(e), & C(e) > 0 \\ 0, & \text{otherwise} \end{cases}`

E_strain is penalty energy, k^(n) is stiffness parameter at Newton iteration n, lambda^(n) is Lagrange multiplier. Applies corrective forces only when excessive stretching occurs, stabilizing cloth dynamics during contact.

Key Figure

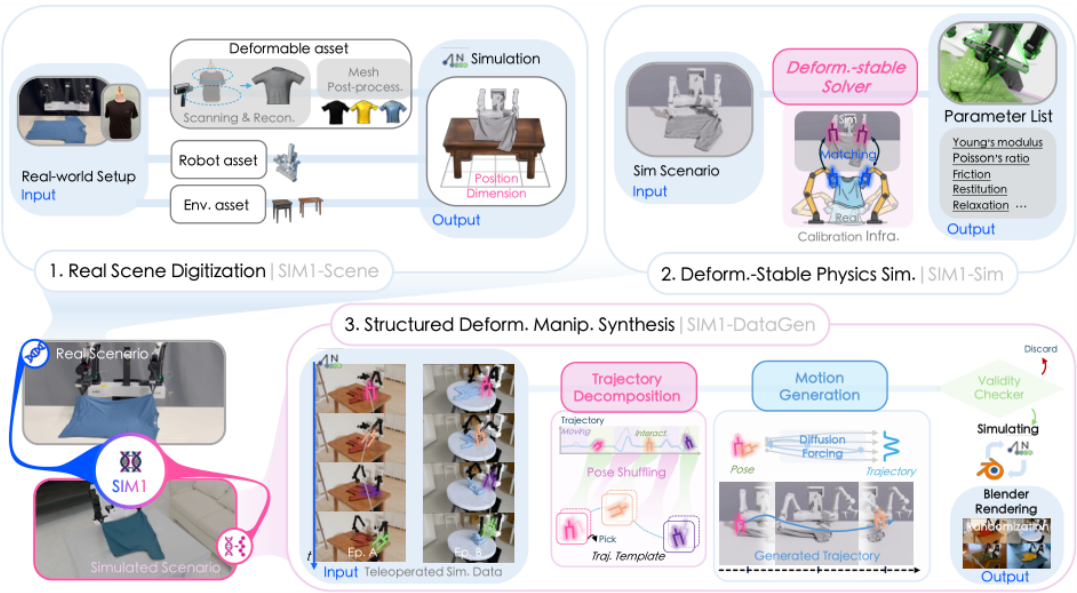

Figure 2. **Framework of SIM1.** **(1)** Real-world objects are reconstructed into metric-accurate, textured simulation assets; **(2)** They are then executed within a deformation-stable simulation framework calibrated through real-to-sim behavior matching. **(3)** Upon physical alignment, diverse manipulation trajectories are synthesized via structured subtask decomposition and diffusion-based motion generation, and rendered with appearance randomization to produce real-equivalent synthetic training data.

Figure 2: Shows the complete SIM1 framework pipeline from real-world digitization through physics-stable simulation to synthetic data generation with visual randomization.

Limitations

- Material calibration requires expert-guided parameter tuning for each asset, constraining automation across arbitrary cloth types
- Real-world validation limited to T-shirt folding task, unclear if approach generalizes to other deformable manipulation scenarios
- Physics solver may still exhibit artifacts in complex contact scenarios despite strain constraints
- Scaling to multiple deformable objects or more complex scenes not demonstrated